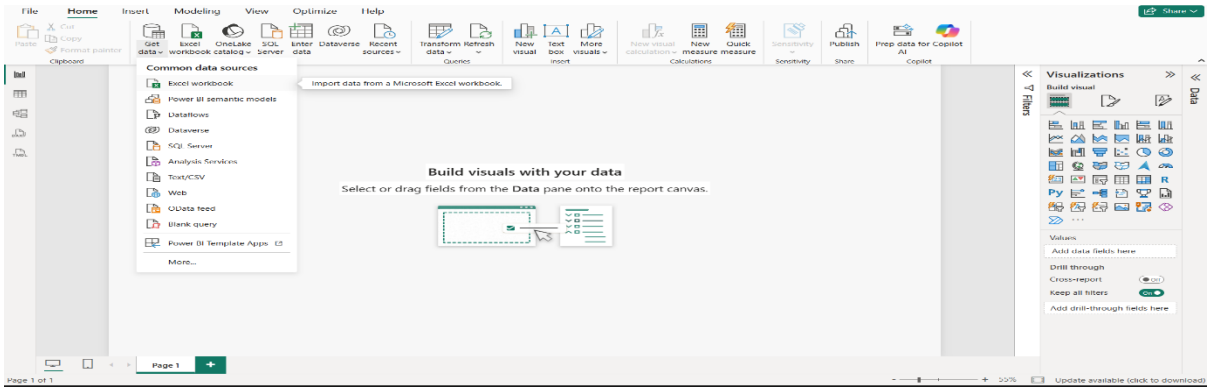
GLOBAL FOOD SECURITY AND SUPPLY CHAIN ANALYSIS

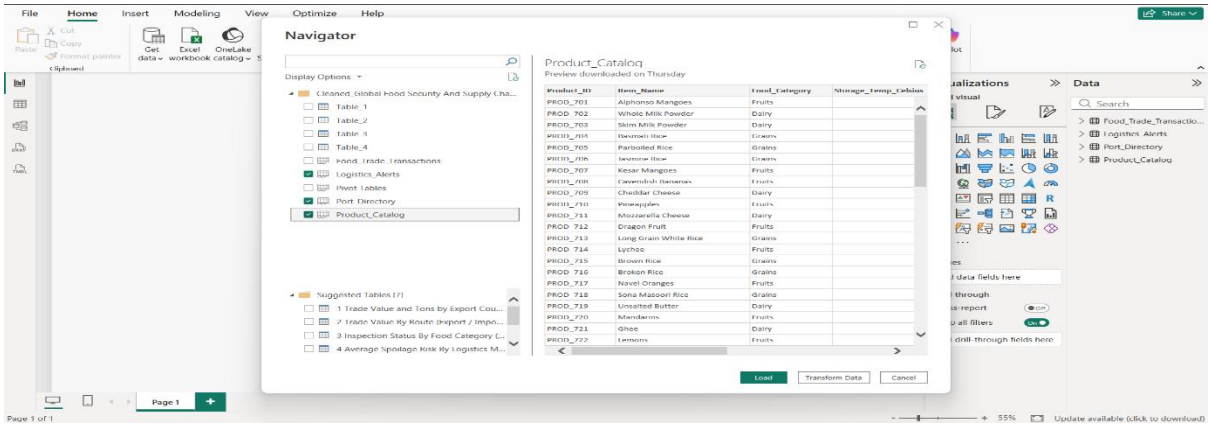
DATA VISUALIZATION USING POWER BI

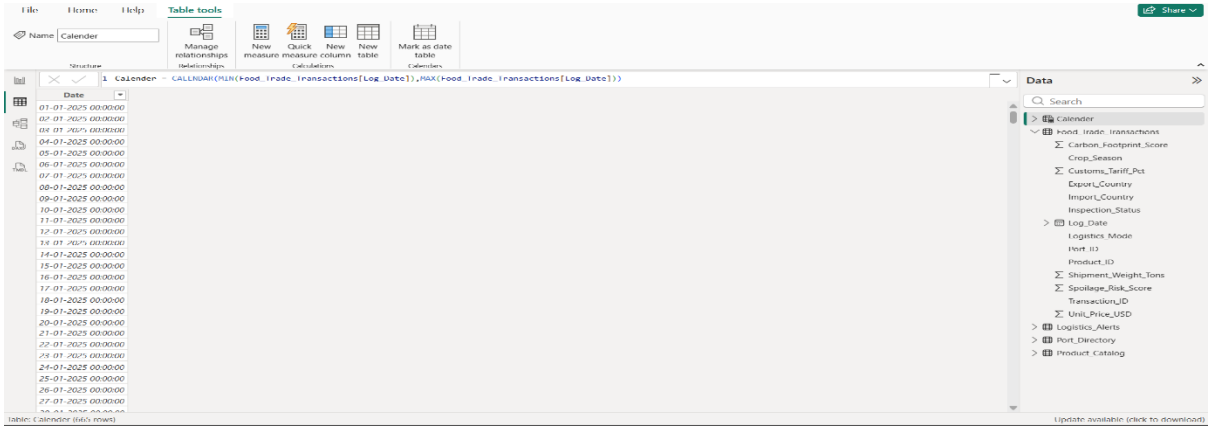
This Project uses 4 Tables

- Food_Trade_Transactions
- Product_Catalog
- Port_Directory
- Logistics_Alerts .

These 4 Tables are Imported And Loaded into the POWER BI .







The new Table is created and are named as CALENDER using the NEW TABLE option and the DAX functions .

The Syntax used :

- Calender = CALENDAR (Min (Food_Trade_Transactions [Log_Date]), MAX (Food_Trade_Transactions [Log_Date]))
- New columns are created using the NEW COLUMN option.
- Year = YEAR(Calender[Date])
- Month = FORMAT(Calender[Date],"MMMM")
- Month_No = MONTH(Calender[Date])
- Quarter = QUARTER(Calender[Date])

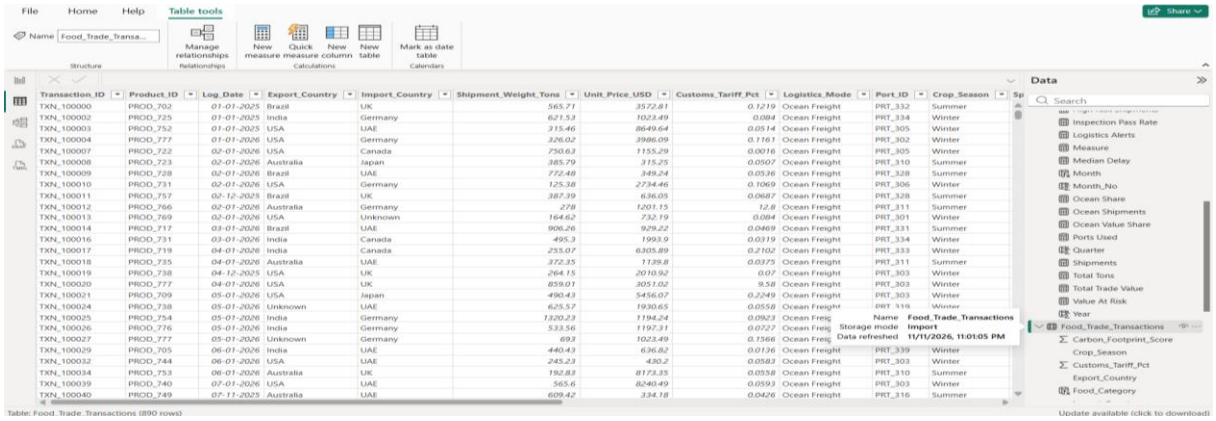


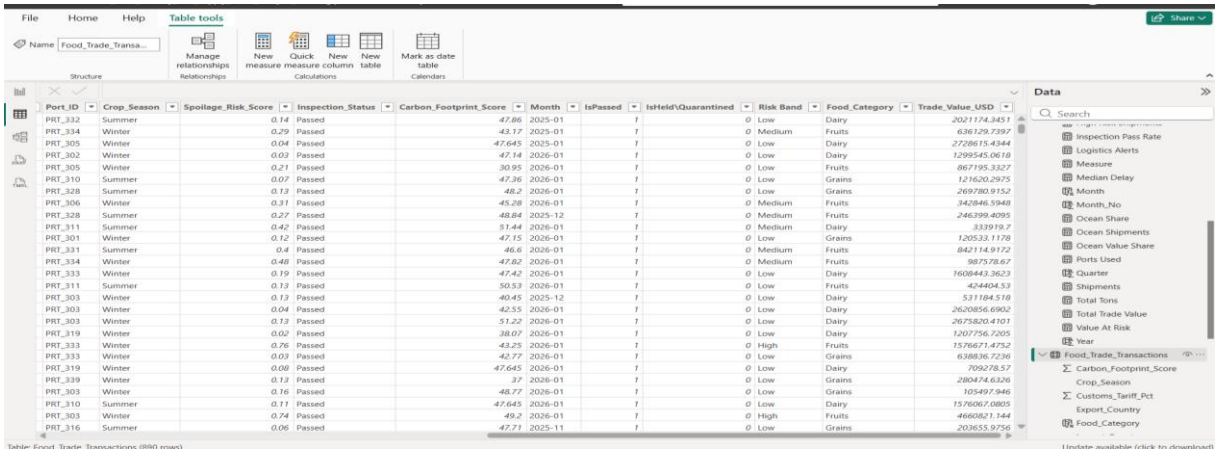
FOOD_TRADE_TRANSACTIONS

This table has 890 rows and 14 columns. Now, 6 new columns are added using the NEW COLUMN option and the DAX functions.

The Syntax Used :

- Food_Category = RELATED(Product_Catalog[Food_Category])
- Trade_Value_USD = Food_Trade_Transactions [Shipment_Weight_Tons] * Food_Trade_Transactions [Unit_Price_USD]
- Month = FORMAT(Food_Trade_Transactions[Log_Date],"YYYY-MM")
- IsPassed = IF(Food_Trade_Transactions[Inspection_Status] = "Passed", 1, 0)
- IsHeld\Quarantined = IF (Food_Trade_Transactions [Inspection_Status] ="Held" || Food_Trade_Transactions [Inspection_Status] = "Quarantined", 1, 0)
- Risk Band = IF (Food_Trade_transactions [Spoilage_Risk_Score] > 0.5, "High", IF (Food_Trade_transactions [Spoilage_Risk_Score] > 0.25, "Medium", "Low"))



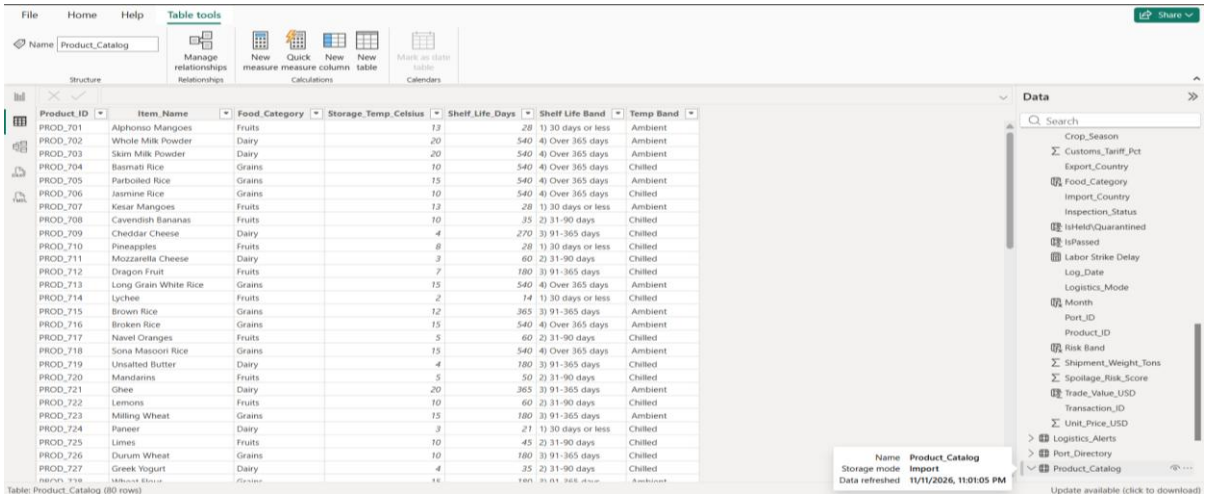


PRODUCT_CATALOG

This table has 80 rows and 5 columns. Now , 2 columns are added using the NEW COLUMN option and the DAX functions.

The Syntax Used :

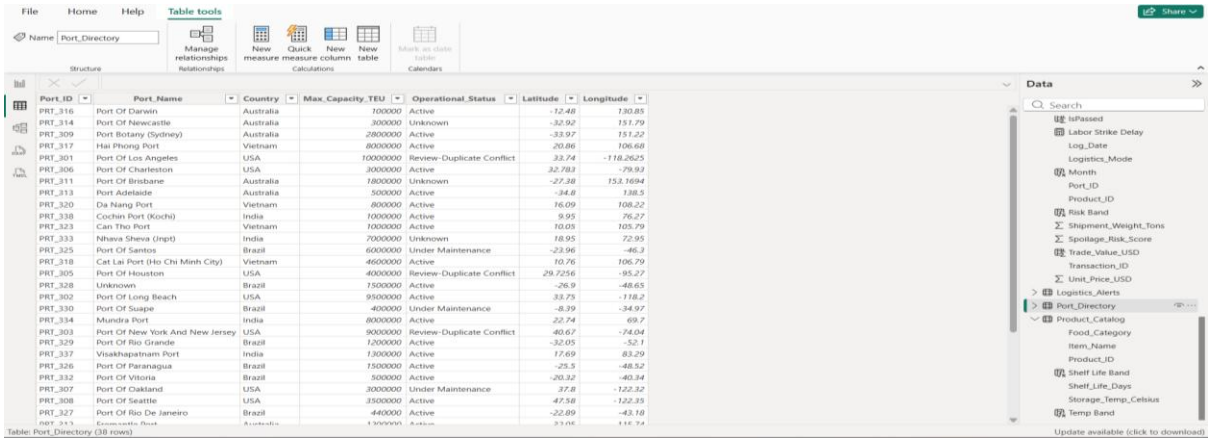
Shelf Life Band = IF(Product_Catalog[Shelf_Life_Days] <= 30, "1) 30 days or less",
IF(Product_Catalog[Shelf_Life_Days] <= 90, "2) 31-90 days",
IF(Product_Catalog[Shelf_Life_Days] <= 365, "3) 91-365 days", "4) Over 365 days")))
Temp Band = IF(Product_Catalog[Storage_Temp_Celsius] <= 0, " Frozen",
IF(Product_Catalog[Storage_Temp_Celsius] <= 10,"Chilled"," Ambient"))



Product_ID	Item_Name	Food_Category	Storage_Temp_Celsius	Shelf_Life_Days	Shelf Life Band	Temp Band
PROD_701	Alphonso Mangoes	Fruits	13	28	1) 30 days or less	Ambient
PROD_702	Whole Milk Powder	Dairy	20	540	4) Over 365 days	Ambient
PROD_703	Skim Milk Powder	Dairy	20	540	4) Over 365 days	Ambient
PROD_704	Basmati Rice	Grains	10	540	4) Over 365 days	Chilled
PROD_705	Parboiled Rice	Grains	15	540	4) Over 365 days	Ambient
PROD_706	Jasmine Rice	Grains	10	540	4) Over 365 days	Chilled
PROD_707	Kesar Mangoes	Fruits	13	28	1) 30 days or less	Ambient
PROD_708	Cavendish Bananas	Fruits	10	35	2) 31-90 days	Chilled
PROD_709	Cheddar Cheese	Dairy	4	270	3) 91-365 days	Chilled
PROD_710	Pineapples	Fruits	8	28	1) 30 days or less	Chilled
PROD_711	Mozzarella Cheese	Dairy	3	60	2) 31-90 days	Chilled
PROD_712	Dragon Fruit	Fruits	7	180	3) 91-365 days	Chilled
PROD_713	Long Grain White Rice	Grains	15	540	4) Over 365 days	Ambient
PROD_714	Lychee	Fruits	2	14	1) 30 days or less	Chilled
PROD_715	Brown Rice	Grains	12	365	3) 91-365 days	Ambient
PROD_716	Broken Rice	Grains	15	540	4) Over 365 days	Ambient
PROD_717	Navel Oranges	Fruits	5	60	2) 31-90 days	Chilled
PROD_718	Sona Masoori Rice	Grains	15	540	4) Over 365 days	Ambient
PROD_719	Unsalted Butter	Dairy	4	180	3) 91-365 days	Chilled
PROD_720	Mandarins	Fruits	5	50	2) 31-90 days	Chilled
PROD_721	Ghee	Dairy	20	365	3) 91-365 days	Ambient
PROD_722	Lemons	Fruits	10	60	2) 31-90 days	Chilled
PROD_723	Milling Wheat	Grains	15	180	3) 91-365 days	Ambient
PROD_724	Paneer	Dairy	3	21	1) 30 days or less	Chilled
PROD_725	Limes	Fruits	10	45	2) 31-90 days	Chilled
PROD_726	Durum Wheat	Grains	10	180	3) 91-365 days	Chilled
PROD_727	Greek Yogurt	Dairy	4	35	2) 31-90 days	Chilled

PORT_DIRECTORY

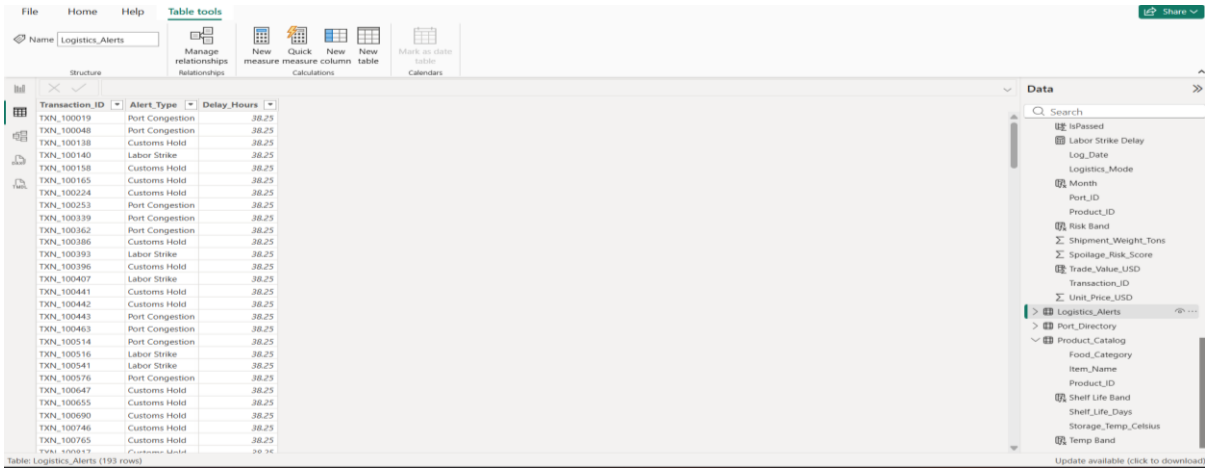
This Table has 38 rows and 7 columns.



Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Status	Latitude	Longitude
PRT_316	Port Of Darwin	Australia	100000	Active	-12.48	130.85
PRT_314	Port Of Newcastle	Australia	300000	Unknown	-32.92	151.79
PRT_309	Port Botany (Sydney)	Australia	2800000	Active	-33.97	151.22
PRT_317	Hai Phong Port	Vietnam	8000000	Active	20.86	106.68
PRT_301	Port Of Los Angeles	USA	10000000	Review-Duplicate Conflict	33.74	-118.2625
PRT_306	Port Of Charleston	USA	3000000	Active	32.783	-79.93
PRT_311	Port Of Brisbane	Australia	1800000	Unknown	-27.38	153.1694
PRT_313	Port Adelaide	Australia	500000	Active	-34.8	138.5
PRT_320	Da Nang Port	Vietnam	800000	Active	16.09	108.22
PRT_338	Cochin Port (Kochi)	India	1000000	Active	9.95	76.27
PRT_323	Can Tho Port	Vietnam	1000000	Active	10.03	105.79
PRT_333	Nhava Sheva (Jnpt)	India	7000000	Unknown	18.95	72.95
PRT_325	Port Of Santos	Brazil	6000000	Under Maintenance	-23.96	-46.3
PRT_318	Cat Lai Port (Ho Chi Minh City)	Vietnam	4600000	Active	10.76	106.79
PRT_305	Port Of Houston	USA	4000000	Review-Duplicate Conflict	29.7256	-95.27
PRT_328	Unknown	Brazil	1500000	Active	-26.9	-48.65
PRT_302	Port Of Long Beach	USA	9500000	Active	33.75	-118.2
PRT_330	Port Of Suape	Brazil	400000	Under Maintenance	-8.39	-34.97
PRT_334	Mundra Port	India	8000000	Active	22.74	69.7
PRT_303	Port Of New York And New Jersey	USA	9000000	Review-Duplicate Conflict	40.67	-74.04
PRT_329	Port Of Rio Grande	Brazil	1200000	Active	-32.05	-52.1
PRT_337	Visakhapatnam Port	India	1300000	Active	17.69	83.29
PRT_326	Port Of Paranagua	Brazil	1500000	Active	-25.5	-48.52
PRT_332	Port Of Vitoria	Brazil	500000	Active	-20.32	-40.34
PRT_307	Port Of Oakland	USA	3000000	Under Maintenance	37.8	-122.32
PRT_308	Port Of Seattle	USA	3500000	Active	47.58	-122.35
PRT_327	Port Of Rio De Janeiro	Brazil	440000	Active	-22.89	-43.18

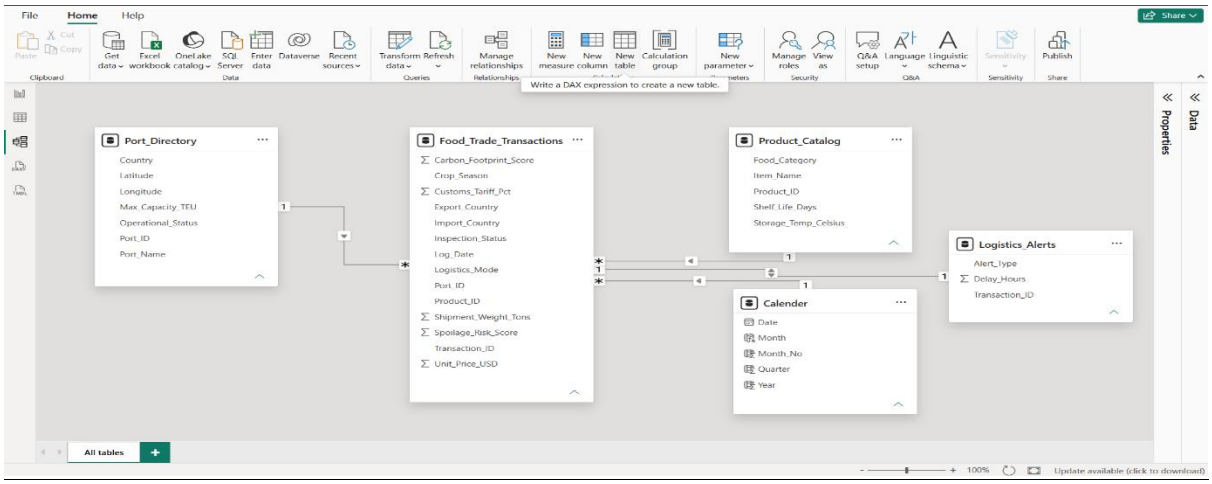
LOGISTICS_ALERTS

This Table has 193 rows and 3 columns.



Transaction_ID	Alert_Type	Delay_Hours
TXN_100019	Port Congestion	38.25
TXN_100048	Port Congestion	38.25
TXN_100138	Customs Hold	38.25
TXN_100140	Labor Strike	38.25
TXN_100158	Customs Hold	38.25
TXN_100165	Customs Hold	38.25
TXN_100224	Customs Hold	38.25
TXN_100253	Port Congestion	38.25
TXN_100339	Port Congestion	38.25
TXN_100362	Port Congestion	38.25
TXN_100386	Customs Hold	38.25
TXN_100393	Labor Strike	38.25
TXN_100396	Customs Hold	38.25
TXN_100407	Labor Strike	38.25
TXN_100441	Customs Hold	38.25
TXN_100442	Customs Hold	38.25
TXN_100443	Port Congestion	38.25
TXN_100463	Port Congestion	38.25
TXN_100514	Port Congestion	38.25
TXN_100516	Labor Strike	38.25
TXN_100541	Labor Strike	38.25
TXN_100576	Port Congestion	38.25
TXN_100647	Customs Hold	38.25
TXN_100655	Customs Hold	38.25
TXN_100690	Customs Hold	38.25
TXN_100746	Customs Hold	38.25
TXN_100765	Customs Hold	38.25

DATA MODELING



Food_Trade_Transactions (Port_ID)----Port_Directory (Port_ID)

Cardinality----Many to One Cross Filter Direction----Single

Food_Trade_Transactions (Product_ID)----Product_Catalog (Product_ID)

Cardinality----Many to One Cross Filter Direction----Single

Food_Trade_Transactions (Transaction_ID)----Logictics_Alerts (Transaction_ID)

Cardinality----One to One Cross Filter Direction----Both

Food_Trade_Transactions (Log_Date)----Calender (Date)

Cardinality----Many to One Cross Filter Direction----Single

REPORTS AND CHARTS .

Here , Cards and Gauges are Created using the NEW MEASURE option and the DAX functions.

REPORT -1

The Syntax Used :

- Total Trade Value = SUM(Food_Trade_Transactions[Trade_Value_USD])
- Shipments = COUNT(Food_Trade_Transactions[Shipment_Weight_Tons])
- Total Tons = SUM(Food_Trade_Transactions[Shipment_Weight_Tons])
- Avg Tariff = AVERAGE (Food_Trade_Transactions [Customs_Tariff_Pct])
- Avg Per Shipment = AVERAGE(Food_Trade_Transactions[Trade_Value_USD])
- Avg Unit Price = AVERAGE(Food_Trade_Transactions[Unit_Price_USD])
- Inspection Pass Rate = AVERAGE(Food_Trade_Transactions[IsPassed])
  MIN : 0 MAX : 100 % TARGET : 95 %
- Avg Spoilage Risk = AVERAGE(Food_Trade_Transactions[Spoilage_Risk_Score])
  MIN : 0 MAX : 50 % TARGET : 30 %

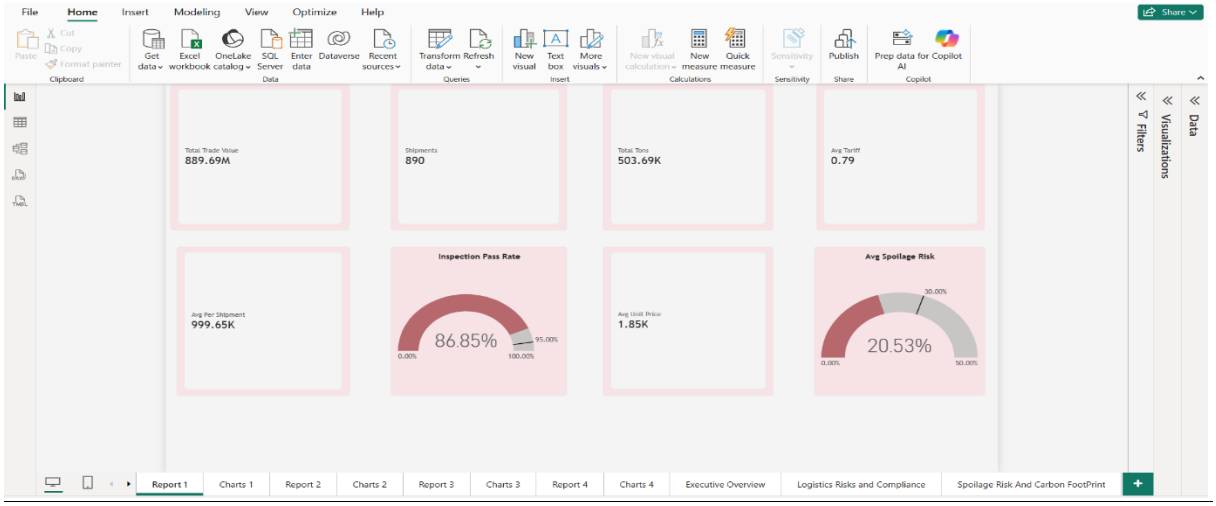
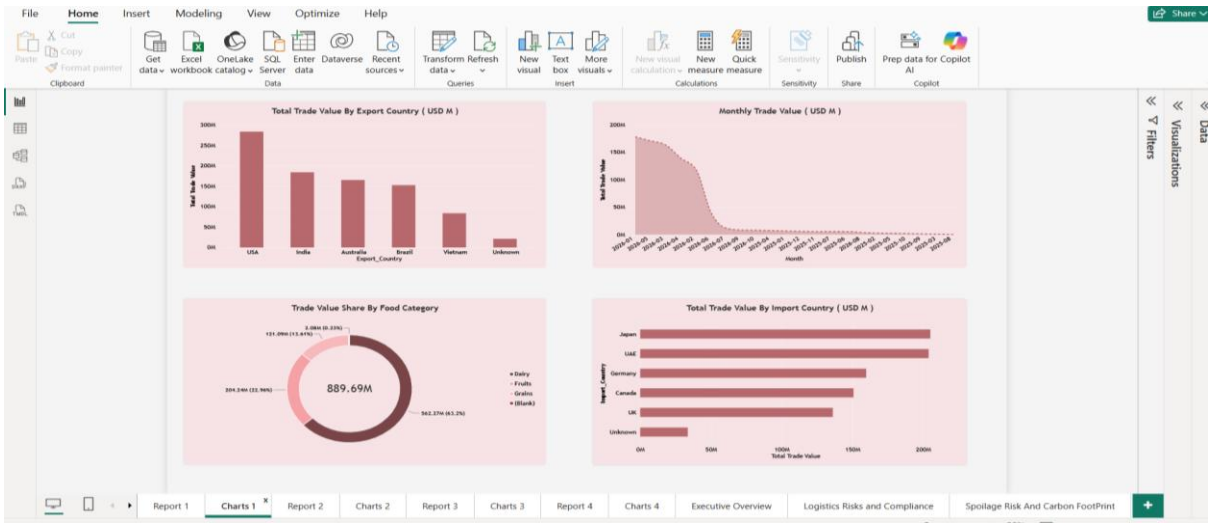


CHART-1

Here, 4 Charts are Used.

- Column Chart
- Bar Chart
- Donut
- Area Chart



REPORT-2

The Syntax Used :

- Logistics Alerts = COUNT(Logistics_Alerts[Alert_Type])
- Median Delay = MEDIAN(Logistics_Alerts[Delay_Hours])
- Ports Used = DISTINCTCOUNT(Port_Directory[Port_ID])
- Held/Quarantined = SUM(Food_Trade_Transactions[IsHeld\Quarantined])
- Active Ports = COUNTROWS(FILTER(Port_Directory, Port_Directory [Operational_Status] = "Active"))
- Ocean Value Share = DIVIDE (CALCULATE (SUM (Food_Trade_Transactions [Trade_Value_USD]),Food_Trade_Transactions[Logistics_Mode] = "Ocean Freight"), SUM(Food_Trade_Transactions[Trade_Value_USD]))
- Average Alert Delay = AVERAGE(Logistics_Alerts[Delay_Hours])
  MIN : 0 MAX : 72 h TARGET : 24h

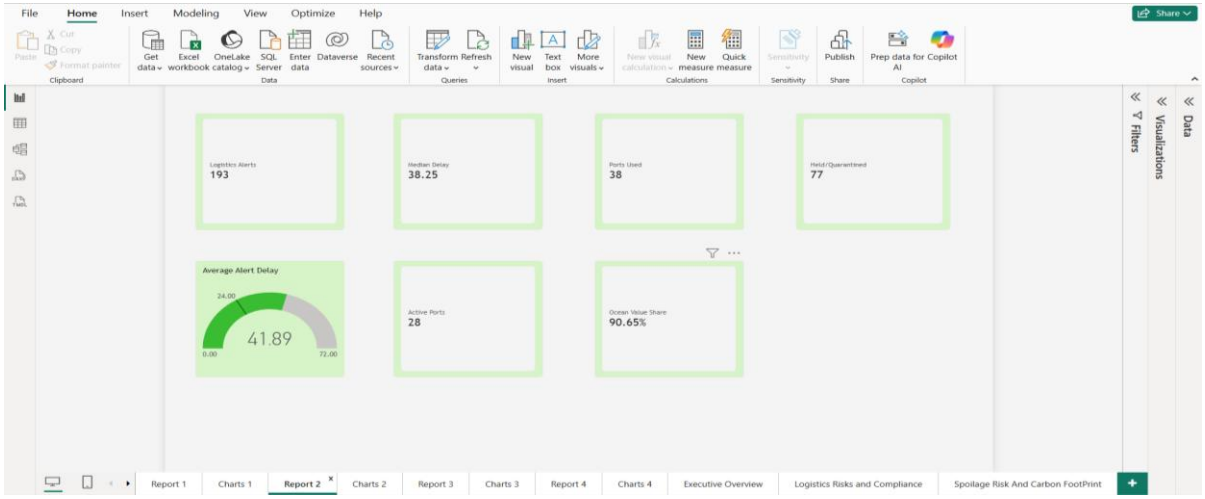
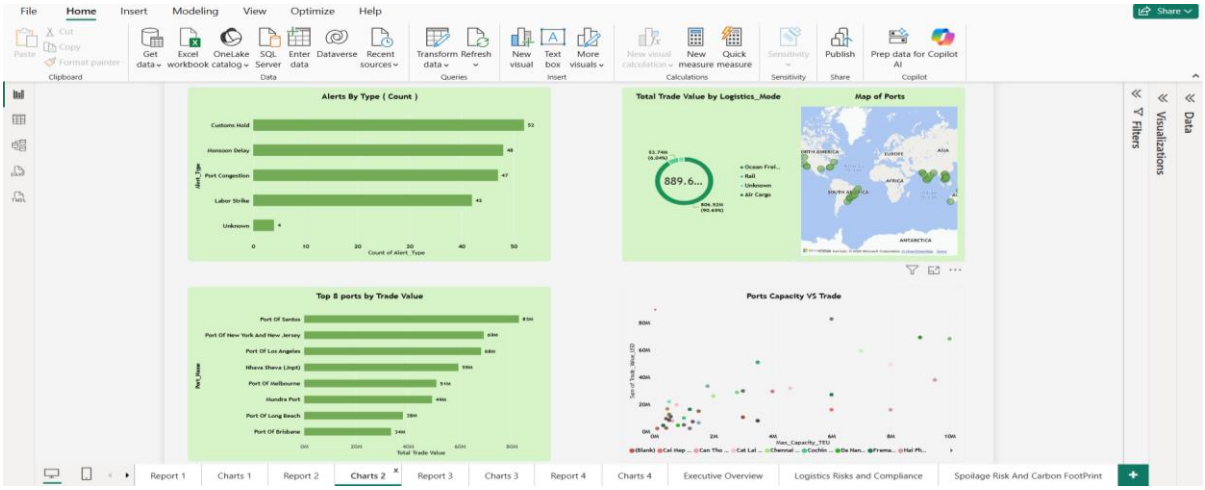


CHART -2

Here, 5 Charts are Used.

- Column Chart
- Bar Chart
- Donut
- Map
- Scatter Chart



REPORT - 3

The Syntax Used :

- High Risk Shipments = COUNTROWS(FILTER(Food_Trade_Transactions, Food_Trade_Transactions[Spoilage_Risk_Score] > 0.5))
- Value At Risk = CALCULATE(SUM(Food_Trade_Transactions[Trade_Value_USD]), Food_Trade_Transactions[Spoilage_Risk_Score] > 0.5)
- High Risk Fruit = COUNTROWS(FILTER(Food_Trade_Transactions, Food_Trade_Transactions[Spoilage_Risk_Score] > 0.5 && Food_Trade_Transactions[Food_Category] = "Fruits"))
- Air Cargo Carbon = CALCULATE(AVERAGE(Food_Trade_Transactions[Carbon_Footprint_Score]), Food_Trade_Transactions[Logistics_Mode] = "Air Cargo")
- High Risk Share = DIVIDE([High Risk Shipments], [Shipments])
  MIN : 0 MAX : 100 % TARGET : 5 %
- Avg Spoilage = AVERAGE(Food_Trade_Transactions[Spoilage_Risk_Score])
- MIN : 0 MAX : 1 TARGET : 0.05
- Avg Carbon Score = AVERAGE (Food_Trade_Transactions [Carbon _ Footprint _ Score])
  MIN : 0 MAX : 100 TARGET : 50

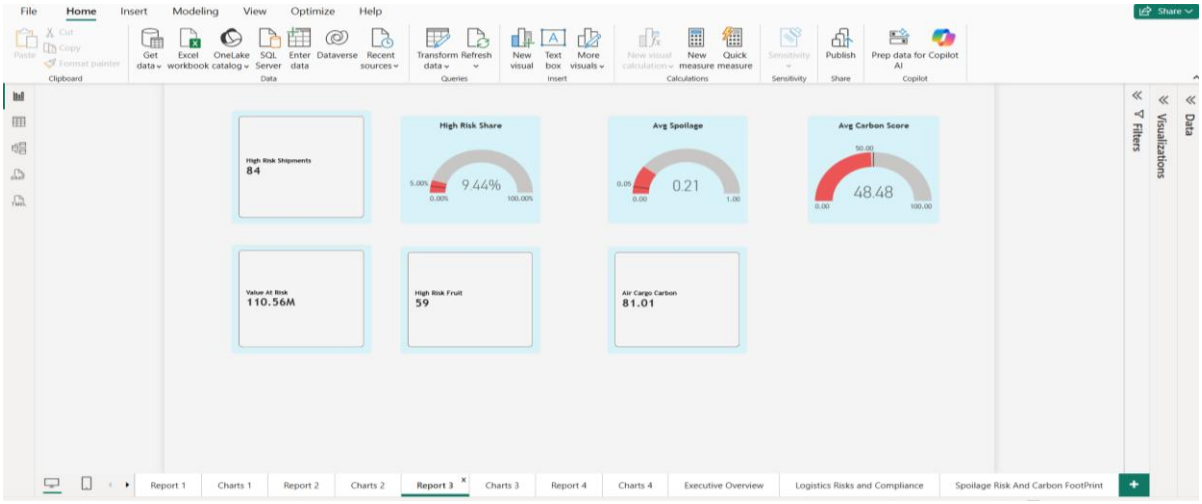
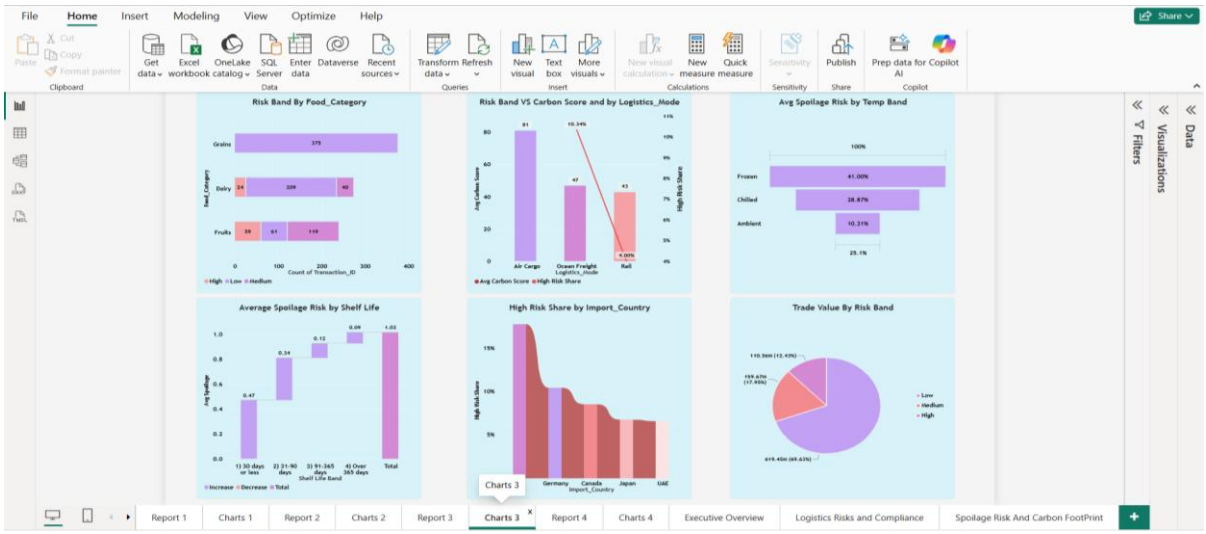


CHART -3

Here, 6 Charts are Used.

- Line and Clustered Column Chart
- Ribbon Chart
- Pie Chart
- Funnel Chart
- Waterfall Chart
- Stacked Bar Chart



REPORT – 4

The Syntax Used :

- Dairy Value = CALCULATE(SUM(Food_Trade_Transactions [Trade_Value_USD]), Food_Trade_Transactions[Food_Category] = "Dairy")
- Dairy Share = DIVIDE([Dairy Value], SUM(Food_Trade_Transactions[Trade_Value_USD]))
- Ocean Share = DIVIDE(COUNTROWS(FILTER(Food_Trade_Transactions, Food_Trade_Transactions[Logistics_Mode] = "Ocean Freight")), COUNTROWS(Food_Trade_Transactions))
- Fruit Spoilage = CALCULATE (AVERAGE (Food_Trade_Transactions [Spoilage_Risk_Score]), Food_Trade_Transactions[Food_Category] = "Fruits")
- Air Cargo Carbon = CALCULATE (AVERAGE (Food_Trade_Transactions [Carbon_Footprint_Score]), Food_Trade_Transactions[Logistics_Mode] = "Air Cargo")
- Labor Strike Delay = CALCULATE(AVERAGE(Logistics_Alerts[Delay_Hours]), Logistics_Alerts[Alert_Type] = "Labor Strike")

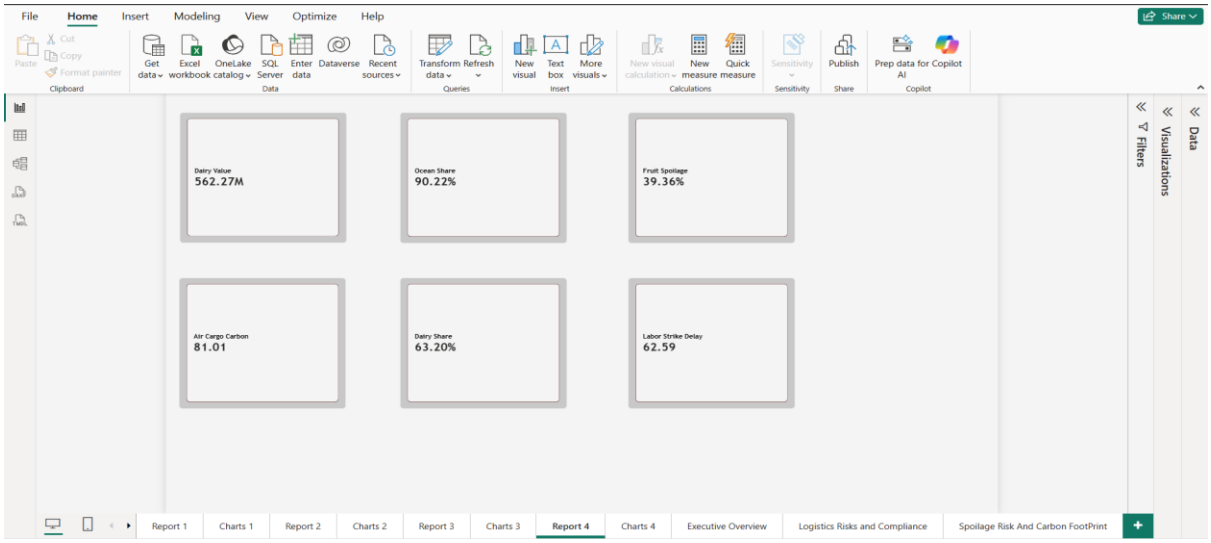
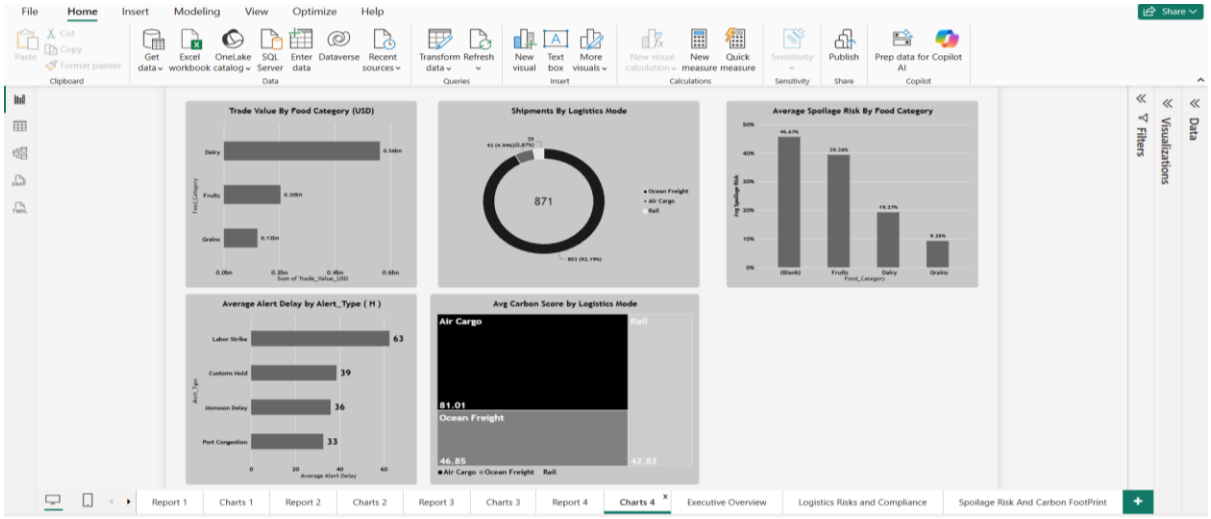


CHART – 4

Here , 4 charts are Used.

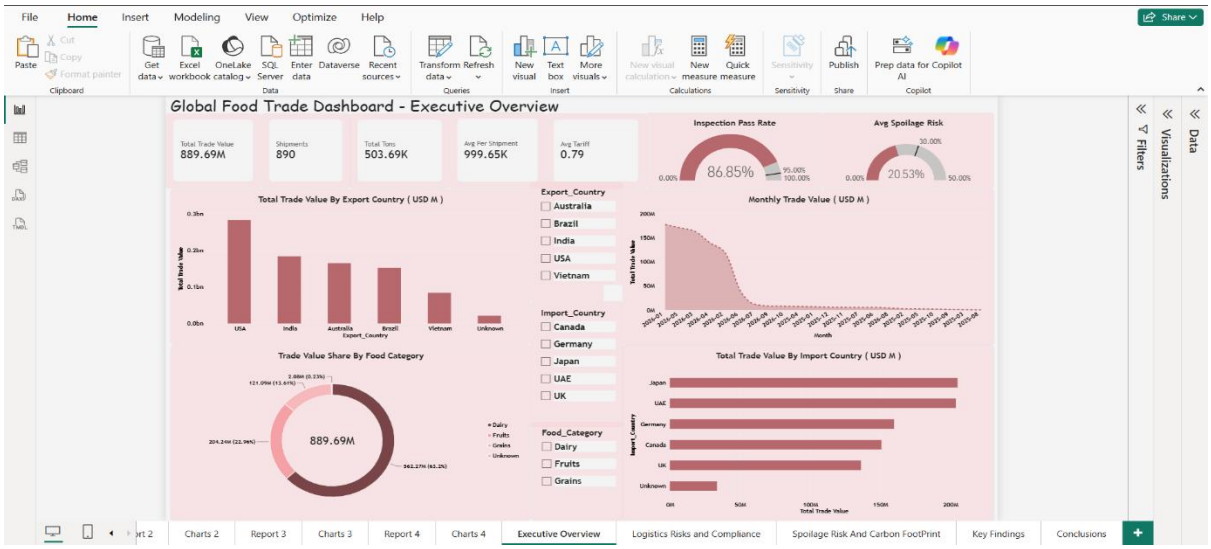
- Clustered Bar Chart
- Clustered Column Chart
- Donut
- Tree Map

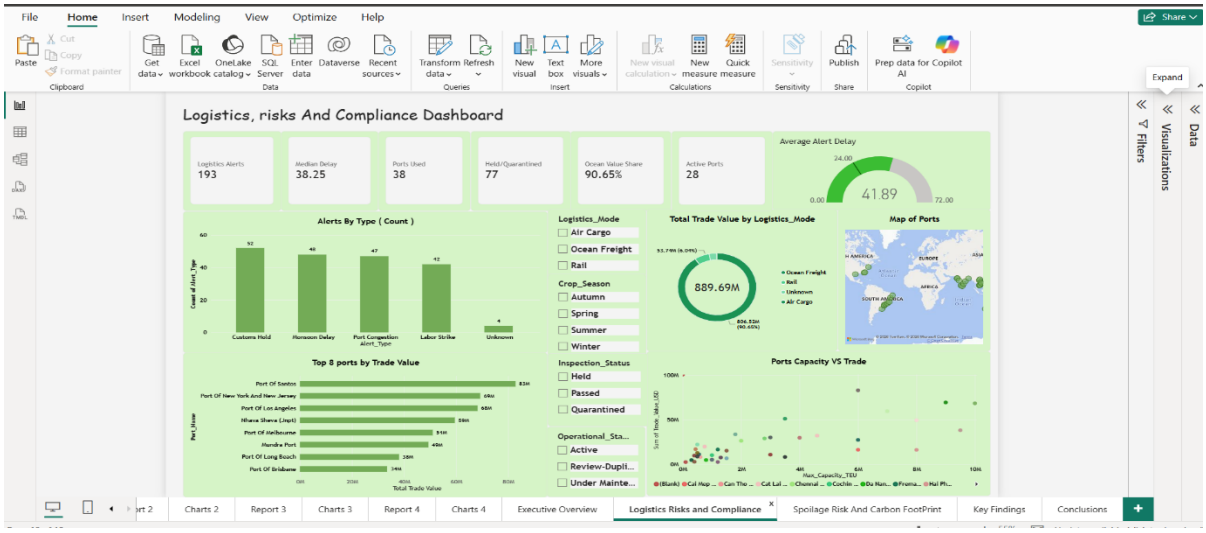


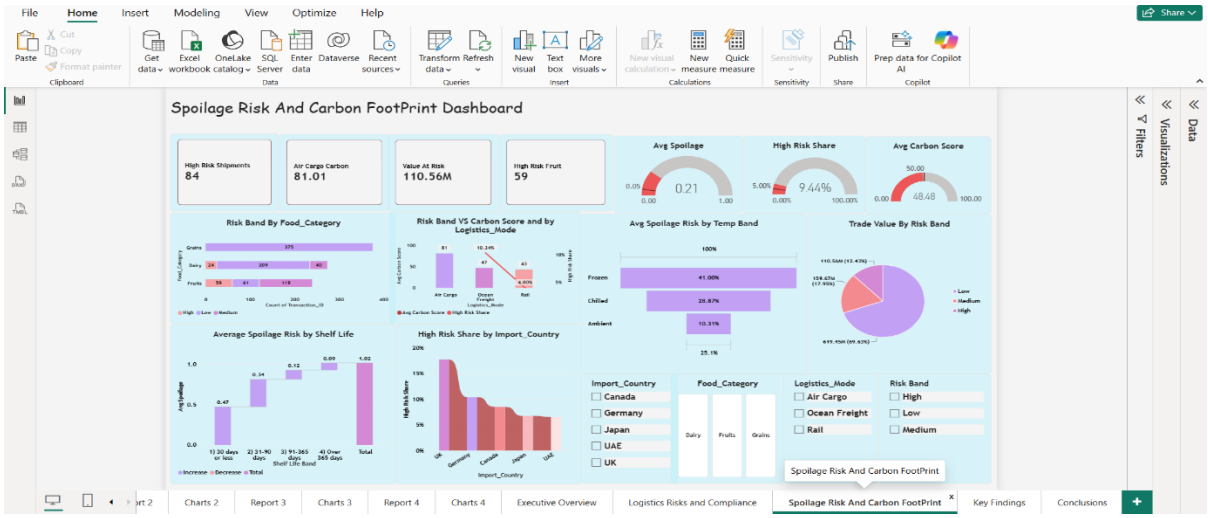
SCREENSHOTS OF DASHBOARDS.

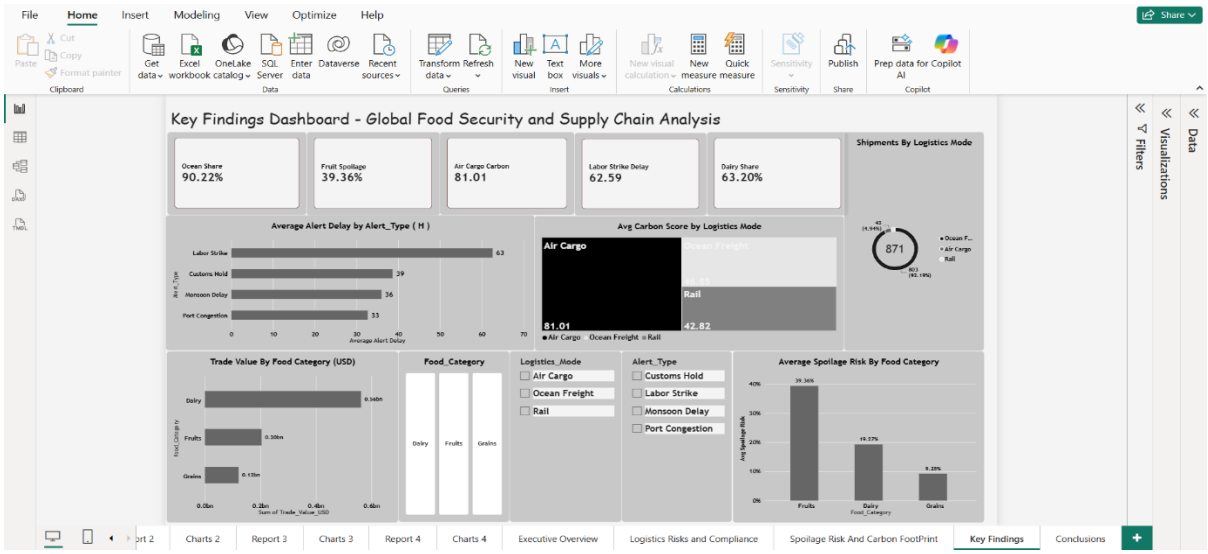
Here , 4 Dashboards are Created.

- Global Food Trade – Executive Overview
- Logistics , Risks and Compliance
- Spoilage Risk and Carbon Footprint
- Key Findings



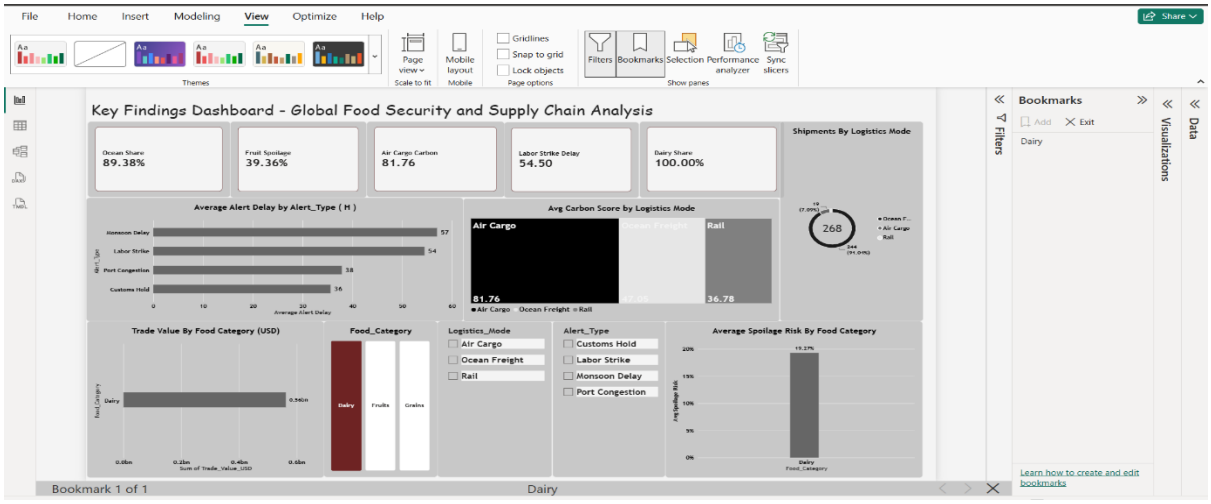

File Home Insert Modeling View Optimize Help
Share
Expand
Logistics, risks And Compliance Dashboard
Logistics Alerts 193
Median Delay 38.25
Ports Used 38
Held/Quarantined 77
Ocean Value Share 90.65%
Active Ports 28
Average Alert Delay
41.89
Alerts By Type (Count)
Customs Hold
Monsoon Delay
Port Congestion
Labor Strike
Unknown
Logistics_Mode
Air Cargo
Ocean Freight
Rail
Crop_Season
Autumn
Spring
Summer
Winter
Inspection_Status
Held
Passed
Quarantined
Operational_Sta...
Active
Review-Dupli...
Under Mainte...
Total Trade Value by Logistics_Mode
889.69M
Map of Ports
Top 8 ports by Trade Value
Ports Capacity VS Trade
Filters
Visualizations
Data
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Key Findings
Conclusions

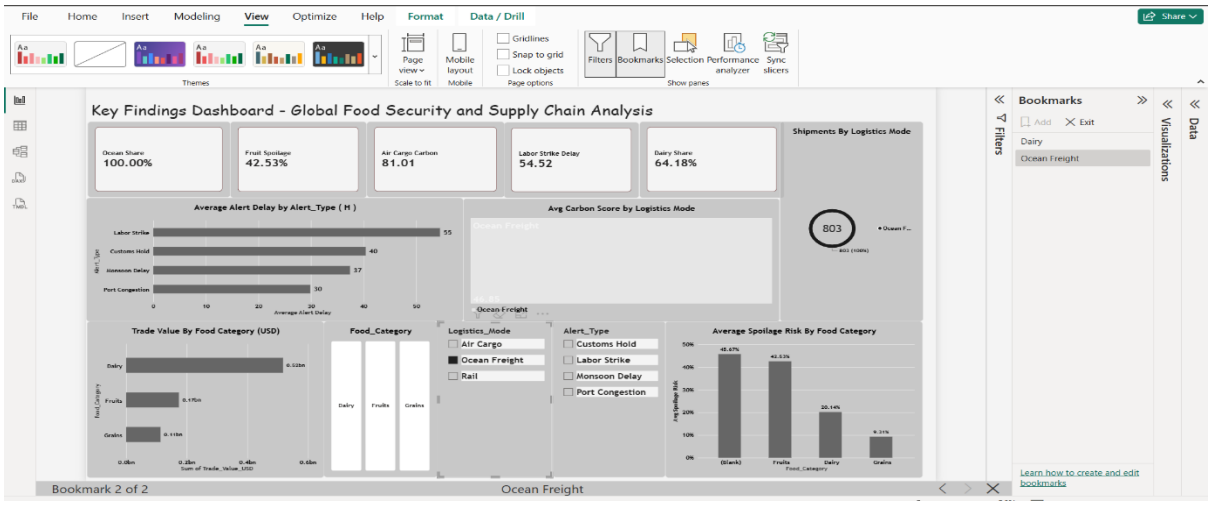

File Home Insert Modeling View Optimize Help
Share
Spoilage Risk And Carbon FootPrint Dashboard
High Risk Shipments 84
Air Cargo Carbon 81.01
Value At Risk 110.56M
High Risk Fruit 59
Avg Spoilage
0.21
High Risk Share
9.44%
Avg Carbon Score
48.48
Risk Band By Food_Category
Risk Band VS Carbon Score and by Logistics_Mode
Avg Spoilage Risk by Temp Band
Frozen
Chilled
Ambient
Trade Value By Risk Band
Average Spoilage Risk by Shelf Life
High Risk Share by Import_Country
Import_Country
Canada
Germany
Japan
UAE
UK
Food_Category
Dairy
Fruits
Grains
Logistics_Mode
Air Cargo
Ocean Freight
Rail
Risk Band
High
Low
Medium
Filters
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Data
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Key Findings
Conclusions

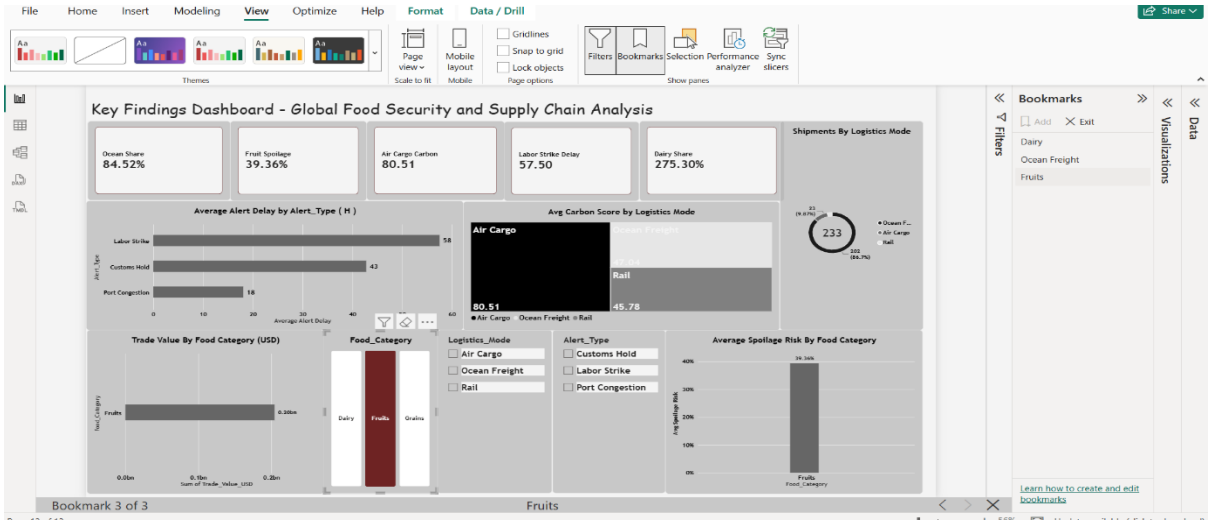

File Home Insert Modeling View Optimize Help
Share
Key Findings Dashboard - Global Food Security and Supply Chain Analysis
Ocean Share 90.22%
Fruit Spoilage 39.36%
Air Cargo Carbon 81.01
Labor Strike Delay 62.59
Dairy Share 63.20%
Shipments By Logistics Mode
871
Average Alert Delay by Alert_Type (H)
Labor Strike
Customs Hold
Monsoon Delay
Port Congestion
Avg Carbon Score by Logistics Mode
Air Cargo
Rail
81.01
42.82
Trade Value By Food Category (USD)
Dairy
Fruits
Grains
Food_Category
Logistics_Mode
Air Cargo
Ocean Freight
Rail
Alert_Type
Customs Hold
Labor Strike
Monsoon Delay
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Average Spoilage Risk By Food Category
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Key Findings
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BOOKMARKS

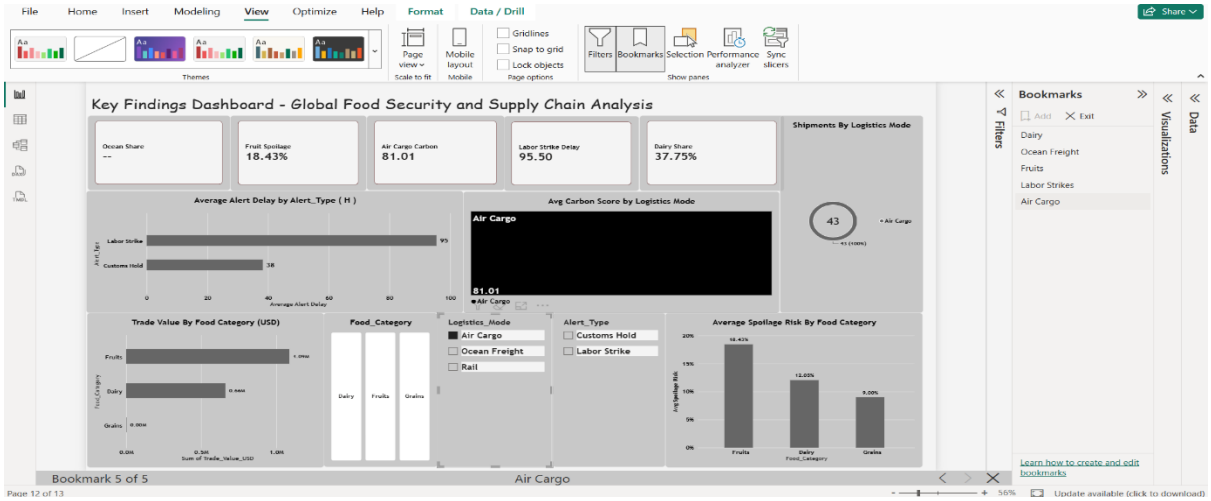
5 Bookmarks are Created to explain the Key Findings.









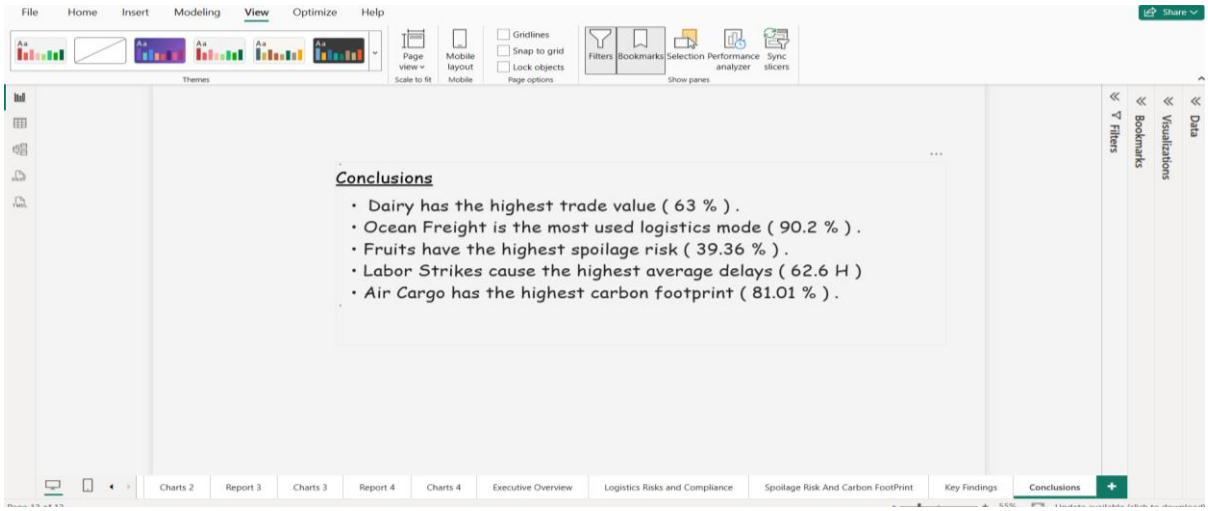


KEY FINDINGS

- Dairy has the highest trade value.
- Ocean Freight is the most used logistics mode.
- Fruits have the highest spoilage risk.
- Labor Strikes cause the highest average delays.
- Air Cargo has the highest carbon footprint.

CONCLUSION

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting. Cleaning the workbook in Excel left 890 valid transactions from 944 rows, and Power BI turned them into interactive dashboards with DAX measures, KPI cards, gauges And slicers .


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Home
Insert
Modeling
View
Optimize
Help
Share
Themes
Page view
Scale to fit
Mobile layout
Mobile
Gridlines
Snap to grid
Lock objects
Page options
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Show panes
Conclusions
• Dairy has the highest trade value (63 %) .
• Ocean Freight is the most used logistics mode (90.2 %) .
• Fruits have the highest spoilage risk (39.36 %) .
• Labor Strikes cause the highest average delays (62.6 H)
• Air Cargo has the highest carbon footprint (81.01 %) .
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